Lab 6

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3. Log Posterior Function

```
x <- c(-1.162, -0.563, -0.091, 0.388, 0.867, 1.389, 1.860, 2.562)
n <- c(603, 1330, 2011, 1556, 646, 289, 145, 61)
z <- c(221, 637, 916, 675, 228, 48, 5, 0)
```

```
log_post <- function(beta, x, n, z) {
  p <- plogis(beta[1] + beta[2] * x)
  loglik <- sum(dbinom(z, size = n, prob = p, log = TRUE))
  logprior <- dnorm(beta[1], mean = 0, sd = 2.5, log = TRUE) +
              dnorm(beta[2], mean = 0, sd = 2.5, log = TRUE)
  loglik + logprior
}
```

```
log_post(c(0, 0), x, n, z)
```

```
[1] -291.0667
```

```
log_post(c(-0.35, -0.37), x, n, z)
```

```
[1] -126.0076
```

```
log_post(c(5, 5), x, n, z)
```

```
[1] -18731.68
```

The log posterior is highest at c(-0.35, -0.37), which makes sense, since beta_0 = -0.35 means the probability is around 0.41 at a 3.8hr transit and declines as transit time increases.

c(0,0) just means p = 0.5 for every bin which is lower, while c(5,5) means predicting almost p = 1 in every bin therefore extremely low.

4. Sampler

```
metropolis <- function(n.iter, prop.sd, x, n, z, beta.init = c(0, 0)) {
  samples <- matrix(NA, nrow = n.iter, ncol = 2)
  colnames(samples) <- c("beta0", "beta1")

  beta.current <- beta.init
  n.accept <- 0

  for (t in 1:n.iter) {
    beta.proposed <- beta.current + rnorm(2, 0, prop.sd)

    log.alpha <- log_post(beta.proposed, x, n, z) -
      log_post(beta.current, x, n, z)

    if (log(runif(1)) < min(0, log.alpha)) {
      beta.current <- beta.proposed
      n.accept <- n.accept + 1
    }

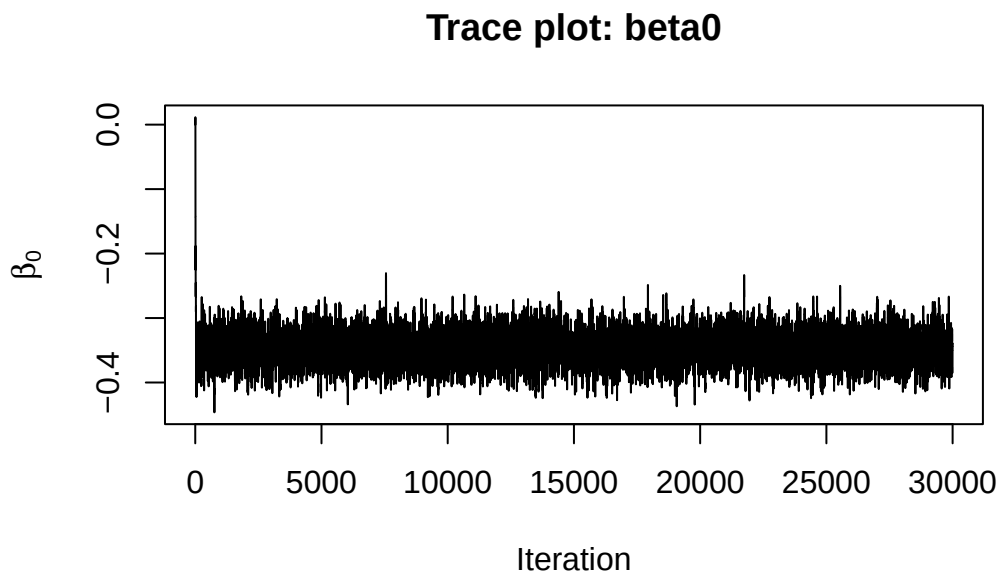
    samples[t, ] <- beta.current
  }

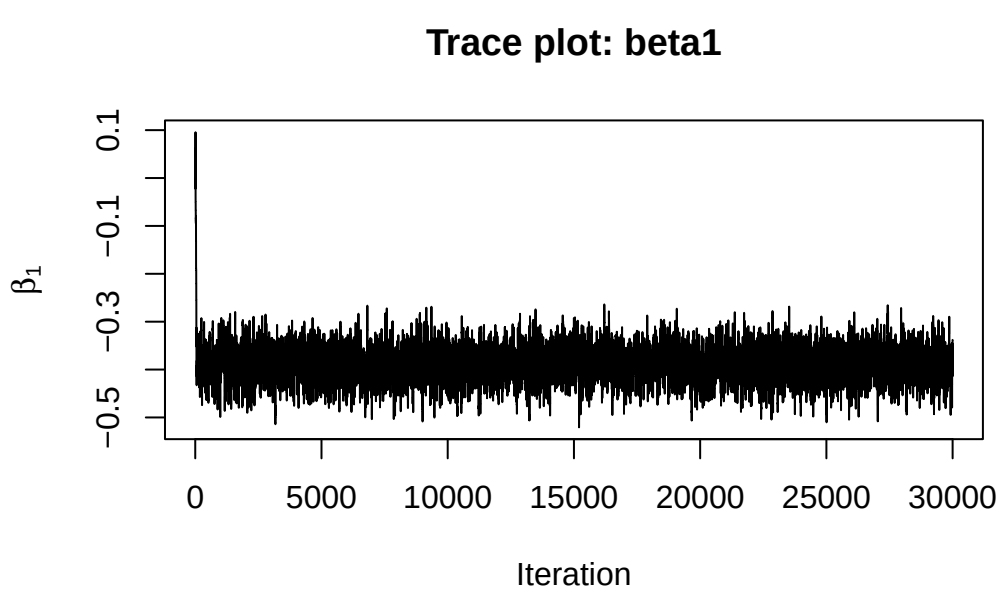
  list(samples = samples, accept.rate = n.accept / n.iter)
}

set.seed(402)
fit <- metropolis(n.iter = 30000, prop.sd = 0.04, x = x, n = n, z = z)
fit$accept.rate
```

```
[1] 0.4468333
```

```
par(mfrow = c(2, 1))
plot(fit$samples[, "beta0"], type = "l", xlab = "Iteration",
     ylab = expression(beta[0]),
     main = "Trace plot: beta0")
plot(fit$samples[, "beta1"], type = "l", xlab = "Iteration",
     ylab = expression(beta[1]),
     main = "Trace plot: beta1")
```






```
par(mfrow = c(1, 1))
```

I started with the prompt prop.sd = 0.04. It gave an acceptance rate of 0.447 and well trace plo9ts so no adjustments were needed.

5. Two starting values.

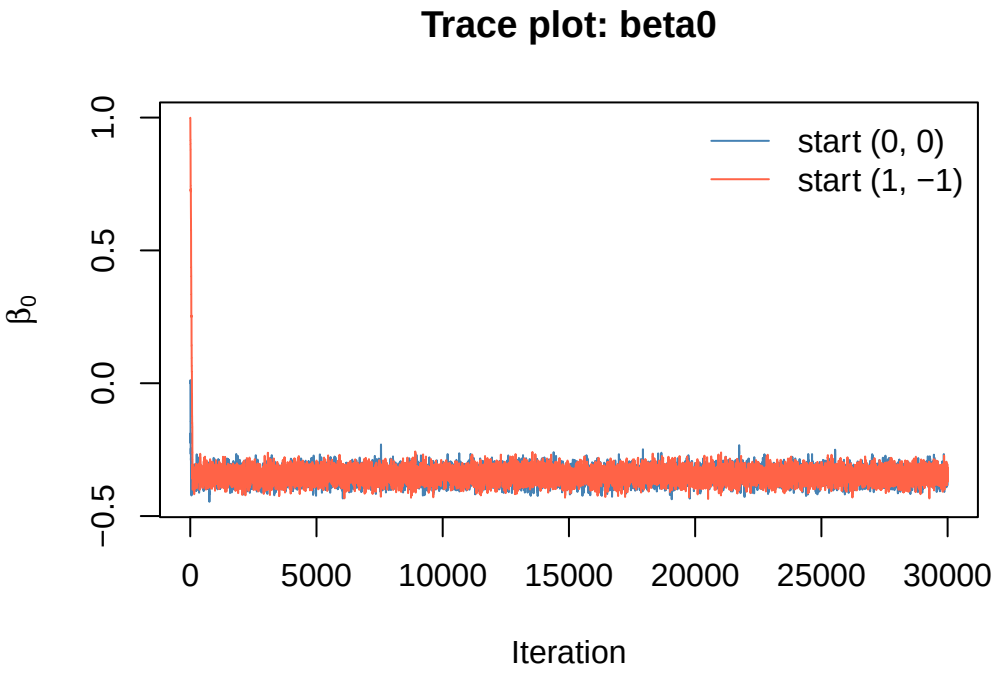
```
fit2 <- metropolis(n.iter = 30000, prop.sd = 0.04, x = x, n = n, z = z,
                   beta.init = c(1, -1))
fit2$accept.rate
```

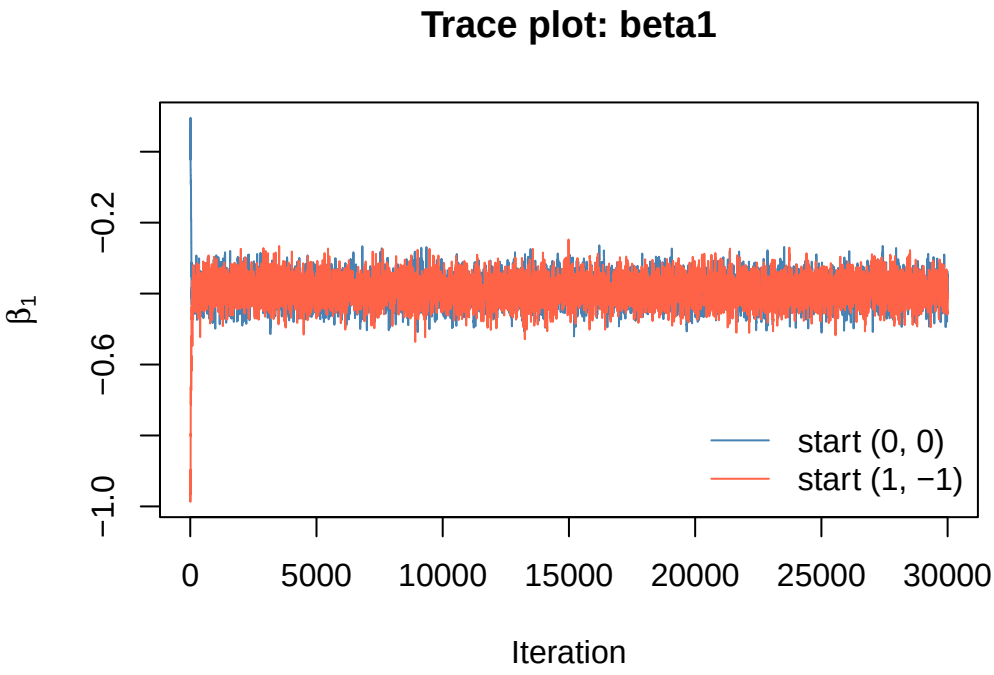
```
[1] 0.4403333
```

```
par(mfrow = c(2, 1))

# beta0
plot(fit$samples[, "beta0"], type = "l", col = "steelblue",
     ylim = range(fit$samples[, "beta0"], fit2$samples[, "beta0"]),
     xlab = "Iteration", ylab = expression(beta[0]),
     main = "Trace plot: beta0")
lines(fit2$samples[, "beta0"], col = "tomato")
legend("topright", legend = c("start (0, 0)", "start (1, -1)"),
       col = c("steelblue", "tomato"), lty = 1, bty = "n")

# beta1
plot(fit$samples[, "beta1"], type = "l", col = "steelblue",
     ylim = range(fit$samples[, "beta1"], fit2$samples[, "beta1"]),
     xlab = "Iteration", ylab = expression(beta[1]),
     main = "Trace plot: beta1")
lines(fit2$samples[, "beta1"], col = "tomato")
legend("bottomright", legend = c("start (0, 0)", "start (1, -1)"),
       col = c("steelblue", "tomato"), lty = 1, bty = "n")
```


Trace plot: beta0
β0
1.0
0.5
0.0
−0.5
start (0, 0)
start (1, −1)
0
5000
10000
15000
20000
25000
30000
Iteration


Trace plot: beta1
β1
−0.2
−0.6
−1.0
start (0, 0)
start (1, −1)
0
5000
10000
15000
20000
25000
30000
Iteration

```r
par(mfrow = c(1, 1))
```

Yes. Both chains move to the same regions within the first few hundred iterations. This is because both have converged to the posterior, and where they started is not important.

6. Retained draws.

```r
warmup <- 1000
draws <- fit$samples[-(1:warmup), ]
nrow(draws)
```

```
[1] 29000
```

I chose a warm up of 1000 iterations to be conservative since both chains seems to be stable after a few hundred iterations. And I keep the remaining 29000 draws for the rest of the analysis.

Part 3: Posterior summaries and prediction

7. Posterior summaries

```r
colMeans(draws)
```

```
     beta0      beta1
-0.3464331 -0.3893759
```

```r
apply(draws, 2, quantile, probs = c(0.025, 0.975))
```

```
           beta0      beta1
2.5%  -0.3959601 -0.4615451
97.5% -0.2963341 -0.3176461
```

The posterior mean of β_0 is -0.35 with a 95% credible interval $(-0.40, -0.30)$.

The posterior mean of β_1 is -0.39 with a 95% credible interval $(-0.46, -0.32)$.

```r
mean(draws[, "beta1"] < 0)
```

```
[1] 1
```

All 29,000 retained draws have $\beta_1 < 0$, so $P(\beta_1 < 0 \mid z, n, x) \approx 1$. This means that we are very certain that Kepler candidates with longer transit duration (more negative slope) are less likely to be confirmed as planets.

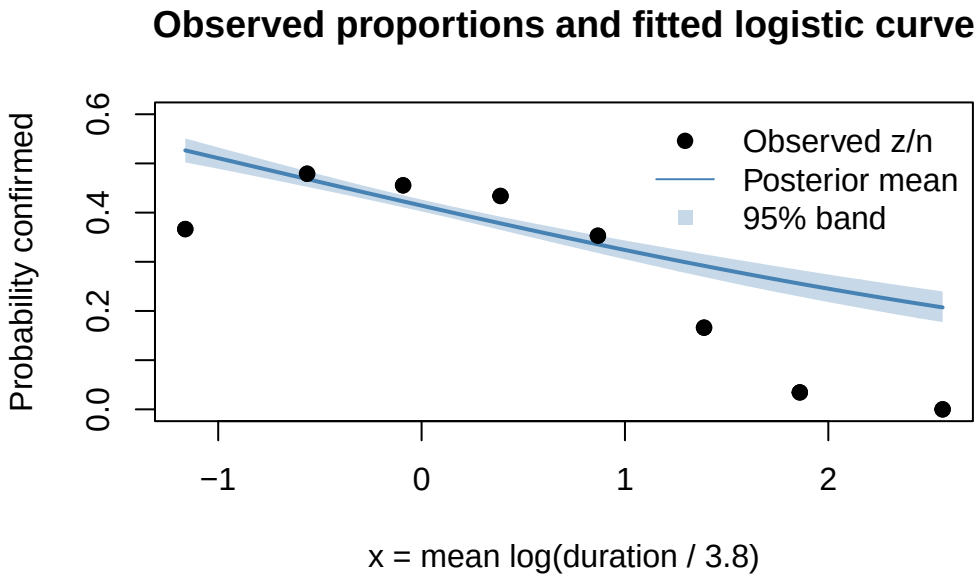
8. A probability curve

```
x.grid <- seq(min(x), max(x), length.out = 200)

# rows = draws, columns = grid values
p.draws <- plogis(outer(draws[, "beta0"], rep(1, length(x.grid))) +
                  outer(draws[, "beta1"], x.grid))

p.mean  <- colMeans(p.draws)
p.lower <- apply(p.draws, 2, quantile, probs = 0.025)
p.upper <- apply(p.draws, 2, quantile, probs = 0.975)
```

```
plot(x, z / n, pch = 19, ylim = c(0, 0.6),
     xlab = "x = mean log(duration / 3.8)", ylab = "Probability confirmed",
     main = "Observed proportions and fitted logistic curve")
polygon(c(x.grid, rev(x.grid)), c(p.lower, rev(p.upper)),
        col = adjustcolor("steelblue", alpha.f = 0.3), border = NA)
lines(x.grid, p.mean, col = "steelblue", lwd = 2)
points(x, z / n, pch = 19)
legend("topright", legend = c("Observed z/n", "Posterior mean", "95% band"),
       pch = c(19, NA, 15), lty = c(NA, 1, NA),
       col = c("black", "steelblue",
               adjustcolor("steelblue", alpha.f = 0.3)), bty = "n")
```



This simple logitic curve does not describe the binned data adequately. There are only two bins that fall in the 95% band. The curve does capture a downward trend, which is valid, but badly over predicts for the last three bins and the first bin.

9. Prediction for a new candidate

$x^* = \log\left(\frac{5}{3.8}\right) = \log(1.3158) \approx 0.274$

```
x.star <- log(5 / 3.8)
p.star <- plogis(draws[, "beta0"] + draws[, "beta1"] * x.star)

mean(p.star)
```

```
[1] 0.3885981
```

```
quantile(p.star, probs = c(0.025, 0.975))
```

```
     2.5%     97.5%
0.3760845 0.4015348
```

The posterior mean is 0.39, with a 95% credible interval (0.376, 0.402).

This intervals says that, given the data and this model, there is a 95% posterior probability that the confirmation probility for candidates with a 5 hour transit lies between 0.38 and 0.40. It does not say whether this particular candidate will be confirmed since the outcome is still very uncertain with a ~39% chance.